



THE UNIVERSITY OF AUCKLAND
NEW ZEALAND

PSEUDONYMISATION AND K-ANONYMITY

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COMPSCI 316



PETs

Pseudonymisation

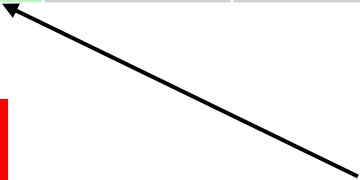
Raw Data

Name	UPI	Age	Gender	Disease
Jhon	dgre6789	29	M	Cancer
Kate	spea6271	22	F	Infection
Alice	ygib9592	35	F	AIDS
Ellen	hbar4405	38	F	AIDS
Mary	mhan6405	41	F	Cancer



Pseudonymised Data

ID	Age	Gender	Disease
P001	29	M	Cancer
P002	22	F	Infection
P003	35	F	AIDS
P004	38	F	AIDS
P005	41	F	Cancer



Pseudonym

ID	Name	UPI
P001	Jhon	dgre6789
P002	Kate	spea6271
P003	Alice	ygib9592
P004	Ellen	hbar4405
P005	Mary	mhan6405

Mapping Table

PETs

Pseudonymisation - Steps

1. Preprocess data
2. Identify direct identifiers
3. Replace direct identifiers with a pseudonym column
4. Create a mapping table **if needed**
5. Secure the mapping table

PETs

Pseudonymisation – Step 2 Generate a Pseudonym Column

- Counter
- Random Number Generator (RNG)
- Cryptographic Hashing
- Message Authentication Code (MAC)
- Encryption

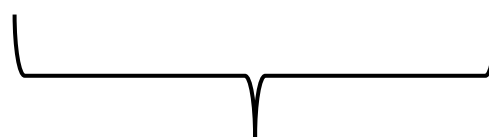
Which one does not need a mapping table?

Let's Code

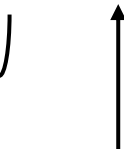
PETs

K-Anonymity

ID	Age	Zip	Gender	Disease
P001	22	2141	M	Cancer
P002	24	2141	F	Infection
P003	31	2138	F	AIDS
P004	32	2139	F	AIDS
P005	41	2243	M	Cancer
P006	41	2245	M	Infection
P007	48	6534	M	Infection



Quasi Identifiers



Sensitive Data



2-Anonymous Data (k=2)

ID	Age	Zip	Gender	Disease
P001	21 – 30	2141	Human	Cancer
P002	21 – 30	2141	Human	Infection
P003	31 – 35	213*	F	AIDS
P004	31 – 35	213*	F	AIDS
P005	41 - 50	*	M	Cancer
P006	41 - 50	*	M	Infection
P007	41 - 50	*	M	Infection

Suppression
Generalisation

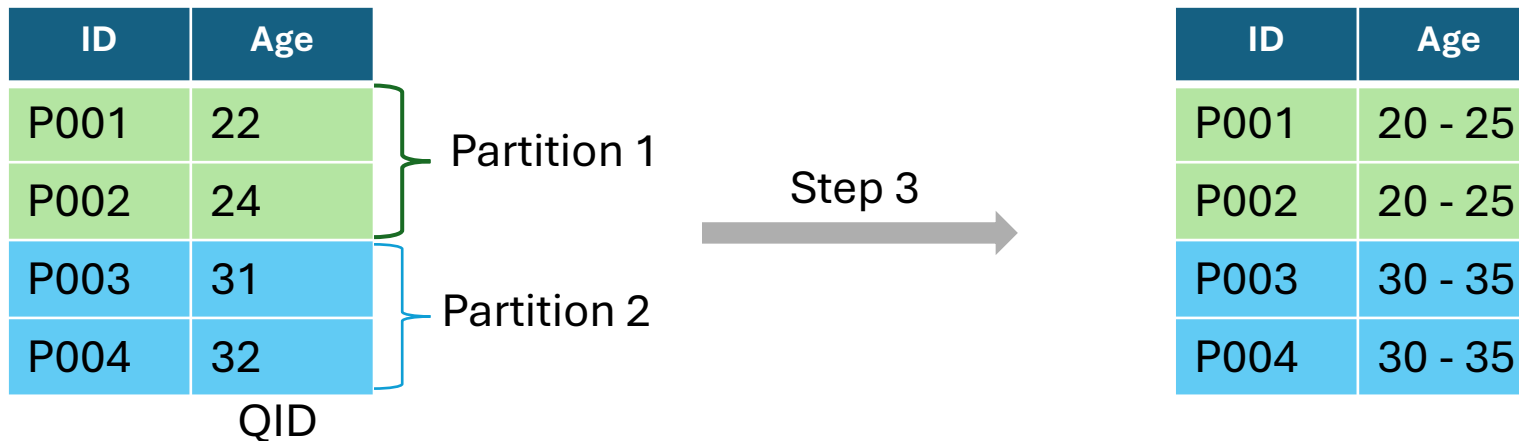
k = 2 => **at least** 2 rows in each group

QIDs {Age, Zip, Gender} in **each group** is the **same**

PETs

K-Anonymity - Steps

1. Preprocess data
2. Identify Quasi Identifiers (QIDs)
3. Partition the dataset : **Each group at least K rows**
4. Anonymise values
For each QID
For each partition



PETs

K-Anonymity - Step 3 Partitioning

How would you partition this for 3-Anonymity?

ID	Age
ID01	24
ID02	23
ID03	21
ID04	20
ID05	22
ID06	24
ID07	20
ID08	21
ID09	22
ID10	23

- Sort 'Age' -> loop -> break into size 3 groups?
- Sort 'Age' -> loop -> break into groups ≥ 3 ?
- What happens when you have multiple QIDs?

We need a fairer way to partition

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Mondrian Algorithm

Sorted Data



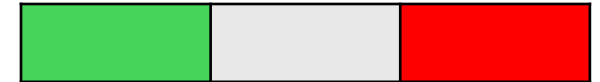
Middle Point



Middle Point



Middle Point



Stop when a condition is met

PETs

K-Anonymity - Step 3 Partitioning (Mondrian Algorithm)

A. Select a partition **P** (at the start : all the rows)

If $\text{size}(\mathbf{P}) \geq 2 * K$

B. Select the most **diverse** QID in the partition (why?)

Find the **median** of the QID values (**M**)

Split the **P** into 2 sub partitions

LHS < **M** and **RHS** $\geq$ **M**

If $\text{size}(\mathbf{LHS}) < K$ OR $\text{size}(\mathbf{RHS}) < K$

If more QIDs exist

Step B for **P**: **remaining** QIDs

else

Add **P** to final_partitions

else

Step A for **P** = **LHS** and **P** = **RHS**

else

Add **P** to final_partitions

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

QIDs

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K-Anonymity - Mondrian Algorithm

Median

1. Numerical Data (e.g., Age)

- Sort
- Find the middle point

2. Categorical Data (e.g., Gender)

- Find the unique values ['M', 'F']
- Divide the unique values into 2 groups

G1 = ['M']

G2 = ['F']

- Match the rows into the correct group

G1 = ['M'] [0,1,2,4,6]

G2 = ['F'] [3,5,7,8,9]

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

QIDs

PETs

Partitioning - Visualise

Assume $K = 3$

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

A. Partition (**P**):

0 1 2 3 4 5 6 7 8 9

If Size (**P**) $\geq 2 * 3$ **true**

B. Select the most **diverse** QID in the partition

Age – 5/10 unique values (**more diverse**)

Gender – 2/10 unique values

qid_variability = ['Age', 'Gender']

Select **'Age'**

Continue....

PETs

Partitioning - Visualise

Assume $K = 3$

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

Find the median (M)

row index	0	1	2	3	4	5	6	7	8	9
Age	24	23	21	20	22	24	20	21	22	23
sorted	3	6	2	7	4	8	1	9	0	5
	20	20	21	21	22	22	23	23	24	24

↑
M = 22

Split the **P** into 2 sub partitions

LHS < 22

3	6	2	7
---	---	---	---

RHS >= 22

4	8	1	9	0	5
---	---	---	---	---	---

Continue....

PETs

Partitioning - Visualise

Assume $K = 3$

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

LHS

3	6	2	7
---	---	---	---

RHS

4	8	1	9	0	5
---	---	---	---	---	---

If size(LHS) < 3 OR size(RHS) < 3 = **false**

Step A for $P = \text{LHS}$ and $P = \text{RHS}$

Continue....

PETs

Partitioning - Visualise

Assume $K = 3$

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

Step A for $P = \text{LHS}$

1

LHS

3 6 2 7

If $\text{size}(\text{LHS}) \geq 2 * 3$

4 $\geq$ 6

false

Add P to final_partitions

3 6 2 7

PETs

Partitioning - Visualise

Assume $K = 3$

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

Step A for $P = \text{LHS}$

1

LHS

3 6 2 7

If $\text{size}(\text{LHS}) \geq 2 * 3$
4 ≥ 6
false

Add P to final_partitions

3 6 2 7

Step A for $P = \text{RHS}$

2

RHS

4 8 1 9 0 5

If $\text{size}(\text{RHS}) \geq 2 * 3$
6 ≥ 6
true

B. Select the most diverse QID in RHS

Age – 3/6 unique values

Gender – 2/6 unique values

qid_vaiability = ['Age', 'Gender']

Select **'Age'**

Continue....

PETs

Partitioning - Visualise

Assume $K = 3$

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

Step A for $P = \text{LHS}$

1

LHS

3 6 2 7

If $\text{size}(\text{LHS}) \geq 2 * 3$
 $4 \geq 6$

false

Add P to final_partitions

3 6 2 7

Step A for $P = \text{RHS}$

2

Find the median (M)

RHS

4 8 1 9 0 5
22 22 23 23 24 24

M = 23

Split the RHS into 2 partitions

LHS < 23

4 8

RHS ≥ 23

1 9 0 5

If $\text{size}(\text{LHS}) < 3$ OR $\text{size}(\text{RHS}) < 3 = \text{true}$

If more QIDs exist = true

Step B on RHS

Continue....

PETs

Partitioning - Visualise

Assume $K = 3$

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

Step A for $P = \text{LHS}$

1

LHS

3 6 2 7

If $\text{size}(\text{LHS}) \geq 2 * 3$

4 $\geq$ 6

false

Add P to final_partitions

3 6 2 7

Step A for $P = \text{RHS}$

2

B. Select the most diverse QID in RHS

qid_variability = [~~'Age'~~, 'Gender']

Select 'Gender'

Find the median (M) of RHS

Continue....

PETs

Partitioning - Visualise

Assume $K = 3$

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

Step A for $P = \text{LHS}$

1

LHS

3 6 2 7

If $\text{size}(\text{LHS}) \geq 2 * 3$
4 ≥ 6
false

Add P to final_partitions

3 6 2 7

Step A for $P = \text{RHS}$

2

Find the median (M)

RHS

4 8 1 9 0 5
M F M F M F

unique_genders = ['M', 'F']

mask_lhs = ['M']

mask_rhs = ['F']

Split the RHS into 2 sub partitions

LHS has mask_lhs RHS has mask_rhs

4 1 0 8 9 5

If $\text{size}(\text{LHS}) < 3$ OR $\text{size}(\text{RHS}) < 3 = \text{false}$

Step A for $P = \text{LHS}$ and $P = \text{RHS}$

Continue....

PETs

Partitioning - Visualise

Assume $K = 3$

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

Step A for $P = \text{LHS}$

1

LHS

3 6 2 7

If $\text{size}(\text{LHS}) \geq 2 * 3$

4 ≥ 6

false

Add P to final_partitions

3 6 2 7

Step A for $P = \text{LHS}$ Step A for $P = \text{RHS}$

2

LHS

4 1 0

If $\text{size}(\text{LHS}) \geq 6$

3 ≥ 6

false

Add LHS to
final_partitions

4 1 0

RHS

8 9 5

If $\text{size}(\text{RHS}) \geq 6$

3 ≥ 6

false

Add LHS to
final_partitions

8 9 5

Stop the Partitioning Process

PETs

K-Anonymity

Assume $K = 3$

final_partitions

3

6

2

7

4

1

0

8

9

5

original

Row Index	ID	Age	Gender
0	ID01	24	M
1	ID02	23	M
2	ID03	21	M
3	ID04	20	F
4	ID05	22	M
5	ID06	24	F
6	ID07	20	M
7	ID08	21	F
8	ID09	22	F
9	ID10	23	F

partitioned

Row Index	ID	Age	Gender
3	ID04	20	F
6	ID07	20	M
2	ID03	21	M
7	ID08	21	F
4	ID05	22	M
1	ID02	23	M
0	ID01	24	M
8	ID09	22	F
9	ID10	23	F
5	ID06	24	F

anonymised

Row Index	ID	Age	Gender
3	ID04	20.5	*
6	ID07	20.5	*
2	ID03	20.5	*
7	ID08	20.5	*
4	ID05	23	M
1	ID02	23	M
0	ID01	23	M
8	ID09	23	F
9	ID10	23	F
5	ID06	23	F

PETs

In the Jupyter Exercise

def mondrian(k)

A. Select a partition **P** (at the start : all the rows)

If size(**P**) >= 2 * **K**

B. Select the most diverse QID in the partition

Find the median of the QID values (**M**)

Split the partition into 2 sub partitions

LHS < **M** and **RHS** >= **M**

If size(**LHS**) < **K** OR size(**RHS**) < **K**

If more QIDs exist

Step B for **P**: for remaining QIDs

else

Add **P** to final_partitions

else

Step A for **P** = **LHS** and **P** = **RHS**

else

Add **P** to final_partitions

def get_qid_variability(partition)

def split(partition, column)

Let's Code



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